

INFO8006: Project 2 - Report

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1 Bayes filter

a. Mathematical Sensor Model of the Rusty Sensor

Let $\mathcal{S} \subset \mathbb{Z}^2$ denote the discrete grid coordinates of traversable maze cells. At time step t , let $X_t = (x_t, y_t) \in \mathcal{S}$ denote the true position of a ghost, and let $P_t = (x_{P,t}, y_{P,t}) \in \mathcal{S}$ denote the known position of Pacman. The true separation is the Manhattan distance:

$$d(X_t, P_t) = \|X_t - P_t\|_1 = |x_t - x_{P,t}| + |y_t - y_{P,t}|. \quad (1)$$

The sensor reading $E_t = e_t \in \mathbb{R}$ is corrupted by zero-mean additive noise W_t :

$$E_t = d(X_t, P_t) + W_t, \quad \text{where } W_t = K_t - np. \quad (2)$$

Here, $K_t \sim \text{Binomial}(n, p)$ is a discrete binomial random variable with parameters $n \in \mathbb{N}_{>0}$ and $p \in (0, 1)$. Since $\mathbb{E}[K_t] = np$ and $\text{Var}(K_t) = np(1 - p)$, the noise W_t satisfies:

$$\mathbb{E}[W_t] = \mathbb{E}[K_t] - np = 0, \quad \text{Var}(W_t) = np(1 - p) = \sigma^2, \quad (3)$$

where σ^2 is the configured sensor variance. In the game configuration, the success probability is fixed to $p = 0.5$, which determines the parameters:

$$\sigma^2 = \frac{n}{4} \implies n = 4\sigma^2, \quad \text{and} \quad np = 2\sigma^2. \quad (4)$$

For the default variance $\sigma^2 = 1.0$, the parameters are $n = 4$ trials and mean shift $np = 2$.

Given a ghost location X_t and Pacman position P_t , the probability of observing measurement e_t is the probability that K_t realizes the integer value $k(e_t, X_t, P_t) = e_t - d(X_t, P_t) + np$:

$$P(E_t = e_t \mid X_t, P_t) = \begin{cases} \binom{n}{e_t - d(X_t, P_t) + np} \left(\frac{1}{2}\right)^n & \text{if } 0 \leq e_t - d(X_t, P_t) + np \leq n, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Consequently, nonzero likelihood is restricted to positions X_t satisfying $d(X_t, P_t) \in [e_t - np, e_t + np]$. For candidate coordinates outside this bounded support, the likelihood is identically zero.

b. Unified Parametric Ghost Transition Model

Let $\mathcal{N}(X_t) \subseteq \mathcal{S}$ denote the accessible non-wall adjacent neighbors of X_t reachable in one step via cardinal movements (North, South, East, West). In Project 2, ghosts cannot stop ($\text{STOP} \notin \mathcal{N}(X_t)$) but are allowed to reverse direction.

Candidate transitions $x' \in \mathcal{N}(X_t)$ are evaluated according to whether they maintain or increase separation from Pacman, $\Delta d(x', X_t, P_t) = d(x', P_t) - d(X_t, P_t) \geq 0$. We introduce a single parameter $c \in [1, \infty)$, called the *fleeing factor*, yielding the unnormalized transition weights:

$$w(x' \mid X_t, P_t; c) = \begin{cases} c & \text{if } d(x', P_t) \geq d(X_t, P_t), \\ 1 & \text{if } d(x', P_t) < d(X_t, P_t). \end{cases} \quad (6)$$

Normalizing over all legal candidate transitions in $\mathcal{N}(X_t)$, the conditional transition distribution is:

$$P(X_{t+1} = x' \mid X_t, P_t) = \begin{cases} \frac{w(x' \mid X_t, P_t; c)}{\sum_{y \in \mathcal{N}(X_t)} w(y \mid X_t, P_t; c)} & \text{if } x' \in \mathcal{N}(X_t), \\ 0 & \text{if } x' \notin \mathcal{N}(X_t). \end{cases} \quad (7)$$

The three ghost behaviors correspond to exact values of $c = 2^k$ for $k \in \{0, 1, 3\}$:

- (i) **Confused Ghost** ($k = 0 \implies c = 2^0 = 1$): $w(x' \mid X_t, P_t; 1) = 1$ for all neighbors. The transition distribution is uniform over legal moves: $P(X_{t+1} = x' \mid X_t, P_t) = \frac{1}{|\mathcal{N}(X_t)|}$. The ghost executes an unbiased random walk, unaware of Pacman.
- (ii) **Afraid Ghost** ($k = 1 \implies c = 2^1 = 2$): Moves preserving or increasing distance have weight 2 versus weight 1 for approaching moves; fleeing moves are twice as likely as approaching ones.
- (iii) **Scared Ghost** ($k = 3 \implies c = 2^3 = 8$): Moves preserving or increasing distance have weight 8 versus weight 1 for approaching moves, driving strong evasive movement away from Pacman.

2 Implementation

- a. *Leave empty.*

3 Experiment

- a. **Belief State Uncertainty Measure: Shannon Entropy**

To summarize the uncertainty in Pacman’s belief state B_t over ghost coordinates, we compute the *Shannon Entropy*:

$$H(B_t) = - \sum_{x \in \mathcal{S}, B_t(x) > 0} B_t(x) \log_2 B_t(x), \quad (8)$$

expressed in bits. Multi-ghost uncertainty is averaged over living ghosts: $\bar{H}_t = \frac{1}{Z} \sum_{i=1}^Z H(B_t^{(i)})$.

Rationale: Shannon entropy is a mathematically rigorous, coordinate-invariant measure of probability dispersion. It attains its theoretical upper bound $H_{\max} = \log_2 |\mathcal{S}|$ under a uniform distribution (complete uncertainty) and its minimum $H = 0$ if and only if all probability mass is concentrated on a single tile (certainty). As the Bayes filter incorporates sensor evidence, entropy tracks the resolution of spatial ambiguity.

- b. **Belief State Quality Measure: Expected Manhattan Distance Error**

Given access to ground-truth ghost coordinates $X_t^* \in \mathcal{S}$, we quantify belief state accuracy via the *Expected Manhattan Distance Error*:

$$\mathcal{E}(B_t, X_t^*) = \mathbb{E}_{X \sim B_t} [\|X - X_t^*\|_1] = \sum_{x \in \mathcal{S}} B_t(x) \cdot \|x - X_t^*\|_1, \quad (9)$$

where $\|x - X_t^*\|_1 = |x_1 - x_1^*| + |x_2 - x_2^*|$ is the grid distance in tiles. Multi-ghost errors are averaged over living ghosts.

Rationale: Standard statistical divergences (such as KL divergence) ignore grid geometry. In contrast, expected Manhattan error directly captures physical proximity in the maze: a belief centered 1 tile away from the ghost incurs an error of ≈ 1.0 tile, while one centered 15 tiles away incurs an error of 15.0 tiles. A value of 0 corresponds to perfect spatial localization.

- c. **Experimental Evaluation on Open and Walled Mazes**

We evaluated the filter across 15 independent runs for each combination of layout (`large_filter`, `large_filter_walls`) and ghost type (`confused`, `afraid`, `scared`) with default variance $\sigma^2 = 1.0$ over $T = 40$ steps. Figure ?? displays mean trajectories with ± 1 standard deviation bands.

Key Findings:

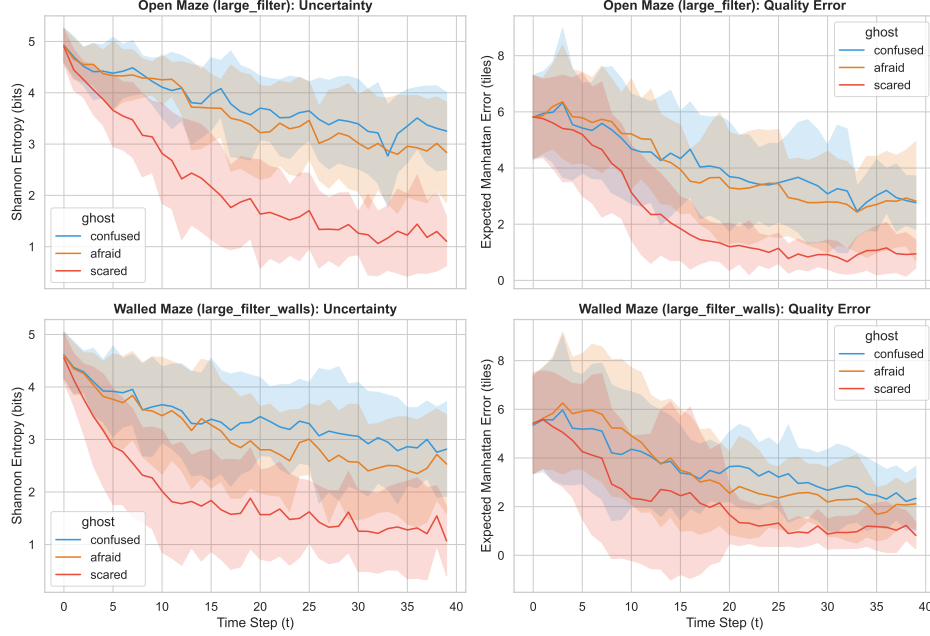


Figure 1: Evolution of uncertainty (Shannon Entropy, left) and quality (Expected Manhattan Error, right) over 40 time steps on the open maze (`large_filter`, top) and the walled maze (`large_filter_walls`, bottom). Solid curves show empirical means across 15 trials; shaded regions depict ± 1 standard deviation.

- **Convergence:** Initial uncertainty and error decline sharply within the first 10–15 steps as noisy distance readings from changing Pacman locations intersect, resolving candidate regions. Both metrics plateau beyond step 25.
- **Ghost Ranking:** Scared ghosts ($c = 8$) yield the lowest steady-state entropy (≈ 1.2 bits) and error (≈ 0.9 tiles). Confused ghosts ($c = 1$) exhibit the highest uncertainty (≈ 3.3 bits on open, ≈ 2.8 bits on walled) and error (≈ 2.8 tiles). Afraid ghosts ($c = 2$) remain strictly intermediate.
- **Consistency:** Variance ribbons narrow significantly as time progresses, demonstrating consistent convergence across seeds.

d. Discussion: Effect of the Ghost Transition Parameter and Maze Topology

Effect on Ghost Behavior: The fleeing factor $c \in \{1, 2, 8\}$ determines the repulsive bias away from Pacman. With $c = 1$ (`confused`), the ghost executes a symmetric random walk, wandering erratically and spreading across hallways. With $c = 8$ (`scared`), the ghost strongly avoids Pacman (moves away are 8 times more probable). In a bounded maze, this pushes the ghost into perimeter corridors, corners, and dead ends. Once cornered, escaping requires moving toward Pacman, which is heavily suppressed by the $\frac{1}{8}$ penalty.

Effect on Belief State: In the prediction step $\bar{B}_{t+1}(x') = \sum_x P(X_{t+1} = x' \mid X_t = x, P_t) B_t(x)$, the transition model acts as a directional prior. When $c = 1$, probability mass is dispersed uniformly in all directions, counteracting sensor contraction. When $c = 8$, probability mass is concentrated in narrow cones pointing away from Pacman, focusing probability into fewer tiles and yielding lower steady-state entropy and localization error.

Maze Topology Comparison: In open space (`large_filter`), scared ghosts migrate to outer boundaries where mobility drops from 4 to 2 directions. In `large_filter_walls`, internal corridors and cul-de-sacs trap scared ghosts; once in a dead end, the ghost cannot escape, collapsing belief into a localized cluster. For confused ghosts, internal walls restrict the average branching factor ($|\mathcal{N}(x)| \leq 3$ vs. 4), slightly dampening spatial diffusion and resulting in lower steady-state entropy on `large_filter_walls` (≈ 2.8 bits) than on `large_filter` (≈ 3.3 bits).

e. Discussion: Effect of Sensor Variance

We investigated sensor noise sensitivity by evaluating $\sigma^2 \in \{0.25, 1.0, 4.0\}$ on `large_filter` against the scared ghost over 15 trials (Figure ??).

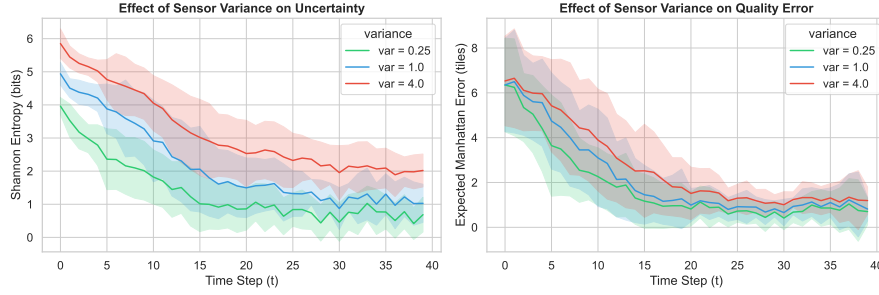


Figure 2: Effect of sensor variance $\sigma^2 \in \{0.25, 1.0, 4.0\}$ on belief state uncertainty (left) and quality error (right) over 40 steps on `large_filter` with scared ghost. Shaded regions depict ± 1 standard deviation.

Theoretical Mechanism: From Section 1.a, $n = 4\sigma^2$ defines binomial noise $W_t = K_t - np$. For $\sigma^2 = 0.25$ ($n = 1$), noise is confined to ± 0.5 tiles; the likelihood function is sharply peaked, eliminating nearly all incompatible tiles in a single update. For $\sigma^2 = 4.0$ ($n = 16$), noise spans up to ± 8 tiles, generating a broad, diffuse likelihood band that provides limited spatial evidence per step.

Empirical Observations:

- **Convergence Rate:** Low variance ($\sigma^2 = 0.25$) reaches < 1.5 bits of entropy within 8 steps, whereas high variance ($\sigma^2 = 4.0$) requires over 25 steps to reach comparable levels.
- **Steady-State Quality:** At $t = 40$, mean entropy scales with noise: ≈ 0.6 bits ($\sigma^2 = 0.25$), ≈ 1.0 bit ($\sigma^2 = 1.0$), and ≈ 2.0 bits ($\sigma^2 = 4.0$). Mean Manhattan error stabilizes at ≈ 0.7 , ≈ 0.9 , and ≈ 1.4 tiles, respectively.
- **Dispersion:** Higher noise introduces broader standard deviation bands, reflecting increased estimation variance across trials.

f. Pacman Controller Strategy for Hunting Ghosts

We designed and implemented a closed-loop controller in `pacmanagent.py` using only permissible inputs: current position P_t , legal actions $\mathcal{A}_{\text{legal}}(P_t)$, and current belief state $B_t = (B_t^{(1)}, \dots, B_t^{(Z)})$. The architecture comprises:

- (1) **Target Selection via Maximum A Posteriori (MAP) Estimation:** For each active ghost i with nonzero belief mass ($\sum_x B_t^{(i)}(x) > 0$), we determine its MAP location:

$$\hat{X}^{(i)} = \arg \max_{x \in S} B_t^{(i)}(x). \quad (10)$$

If multiple ghosts are active, Pacman targets the closest active MAP estimate in Manhattan distance: $i^* = \arg \min_{i:\text{alive}} \|P_t - \hat{X}^{(i)}\|_1$, prioritizing immediate high-confidence targets and avoiding oscillations.

- (2) **Shortest-Path Navigation via Breadth-First Search (BFS):** Using Pacman’s position P_t as root, target $\hat{X}^{(i^*)}$ as goal, and the maze wall topology, a Breadth-First Search (BFS) computes an optimal, collision-free shortest path $\pi^* = (P_t, p_1, \dots, \hat{X}^{(i^*)})$. Pacman extracts the initial action $a^* = \text{action}(P_t \rightarrow p_1) \in \mathcal{A}_{\text{legal}}(P_t)$.
- (3) **Closed-Loop Receding-Horizon Control:** Pacman executes only the first step a^* each cycle. At $t + 1$, the Bayes filter incorporates new sensor evidence, updating the posterior. The controller recomputes the MAP target and replans the BFS path dynamically, tracking and intercepting fleeing ghosts robustly.

g. *Leave empty.*